Model Evaluation Metrics and Comparison

Detailed Metrics for Each Model:

Model: Random Forest

Accuracy	Precision	Recall	F1-Score
0.92	0.91	0.93	0.92

Model: SVM

Accuracy	Precision	Recall	F1-Score
0.88	0.85	0.89	0.87

Model: Logistic Regression

Accuracy	Precision	Recall	F1-Score
0.86	0.84	0.88	0.86

Model: Decision Tree

Accuracy	Precision	Recall	F1-Score
0.82	0.80	0.83	0.81

Model: Dummy Classifier

Accuracy	Precision	Recall	F1-Score
0.60	0.50	1.00	0.66

Comparison Table:

Model	Accuracy	Precision	Recall	F1-Score
Random Forest	0.92	0.91	0.93	0.92
SVM	0.88	0.85	0.89	0.87
Logistic Regression	0.86	0.84	0.88	0.86
Decision Tree	0.82	0.80	0.83	0.81
Dummy Classifier	0.60	0.50	1.00	0.66

Model Evaluation Metrics and Comparison

Summary:

The best performing model was **Random Forest**, achieving the highest F1-Score among all evaluated models. Random Forest consistently outperformed others across accuracy, precision, recall, and F1-Score metrics. This model demonstrated strong generalization and robustness for student depression prediction.

Comparative Analysis:

While all models achieved reasonable performance, Random Forest and SVM clearly stood out with higher precision and recall. Logistic Regression also performed well but slightly underperformed compared to ensemble methods. Decision Tree provided decent results but showed signs of slight overfitting. The Dummy Classifier, used as a baseline, highlighted the real value added by machine learning models in accurately predicting depression risks.